

UNIVERSITI TENAGA NASIONAL

The Energy University

PKIB113

CREATIVE THINKING

Credit: PhonlamaiPhoto *Getty Images*

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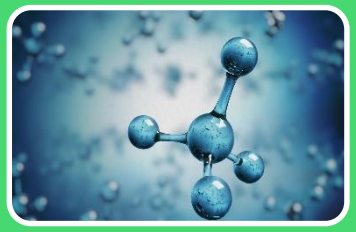
INTRODUCTION TO PKIB113 (CREATIVE THINKING) SPECIAL SEMESTER, 2024/2025

COURSE MAP



INTRODUCTION

- What is creativity?
- Definitions and concepts



KEY PRINCIPLES AND THEORIES IN CREATIVITY

- Creative Thinking elements
- Creative Thinking process



CREATIVE INNOVATIONS

- Idea Generation Techniques
- Brainstorming, mind map, FC, SCAMPER



CREATIVE PROBLEM SOLVING

- De Bono's Thinking Tools:
- PMI, OPV
- C&S, CAF
- Six Thinking Hats

COURSE OUTLINE & ASSESSMENTS

ASSESSMENTS	DESCRIPTION	PERCENTAGE
PORTFOLIO	Padlet + Self-reflection	25%
ASSIGNMENT 1	Creative Design and Innovation	20%
ASSIGNMENT 2	Creative Problem Solving	25%
FINAL EXAMINATION		30%
TOTAL		**100%**

TL + COMMUNICATION CHANNELS

1. BRIGHTEN - LMS
2. CLASS TELEGRAM – Main communication channel: queries, updates, reminders
3. Contact classes – F2F scheduled classes
4. E-mail – absent reports, course work follow-up, carry marks
5. Portal notice – official reminders for classes, exams
6. Whatsapp – quicker response
7. PADLET

TO DO

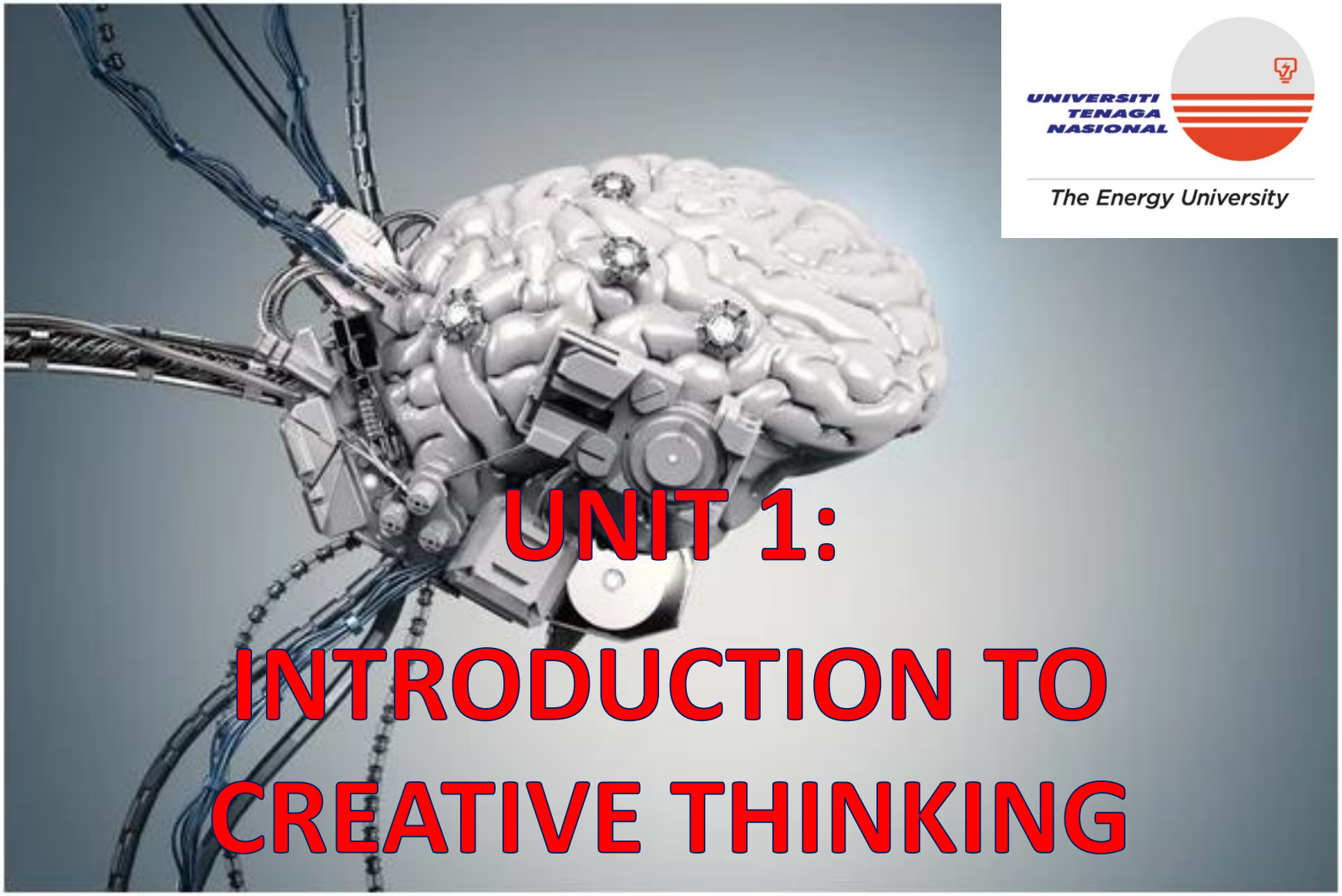




- **Form a group** that is workable for you
- If no group, I will assist at a given deadline

- **Access class PADLET**!

- **Join CLASS TELEGRAM**!



UNIVERSITI TENAGA NASIONAL

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UNIT 1:

INTRODUCTION TO CREATIVE THINKING

Credit: PhonlamaiPhoto *Getty Images*

OVERVIEW OF UNIT 1

1.1 Creativity as a limitless concept

1.2 Value Creation

1.3 Simple and complex definition

1.4 The Big C and the small c


1.1
CREATIVITY AS A LIMITLESS CONCEPT

1.1 CREATIVITY AS A LIMITLESS CONCEPT

- Before we begin the course…
- What comes to your mind when you hear the word “creative”?



Is this creative?

The Toilet House, South Korea



magicbricks.com

Tactical laser pizza cutter



Basketball mug



interestingengineering.com

Is this creative?

The transparent house



archdaily.com



HOW DO WE DEFINE CREATIVITY?

The concept of creativity is:

- open
- infinite
- limitless

So, how do we **DEFINE** it?

HOW DO WE MEASURE CREATIVITY?

The concept of creativity is also:

- subjective
- depends on one's perspective
- depends on one's standard of expectation

So, how do we MEASURE it?



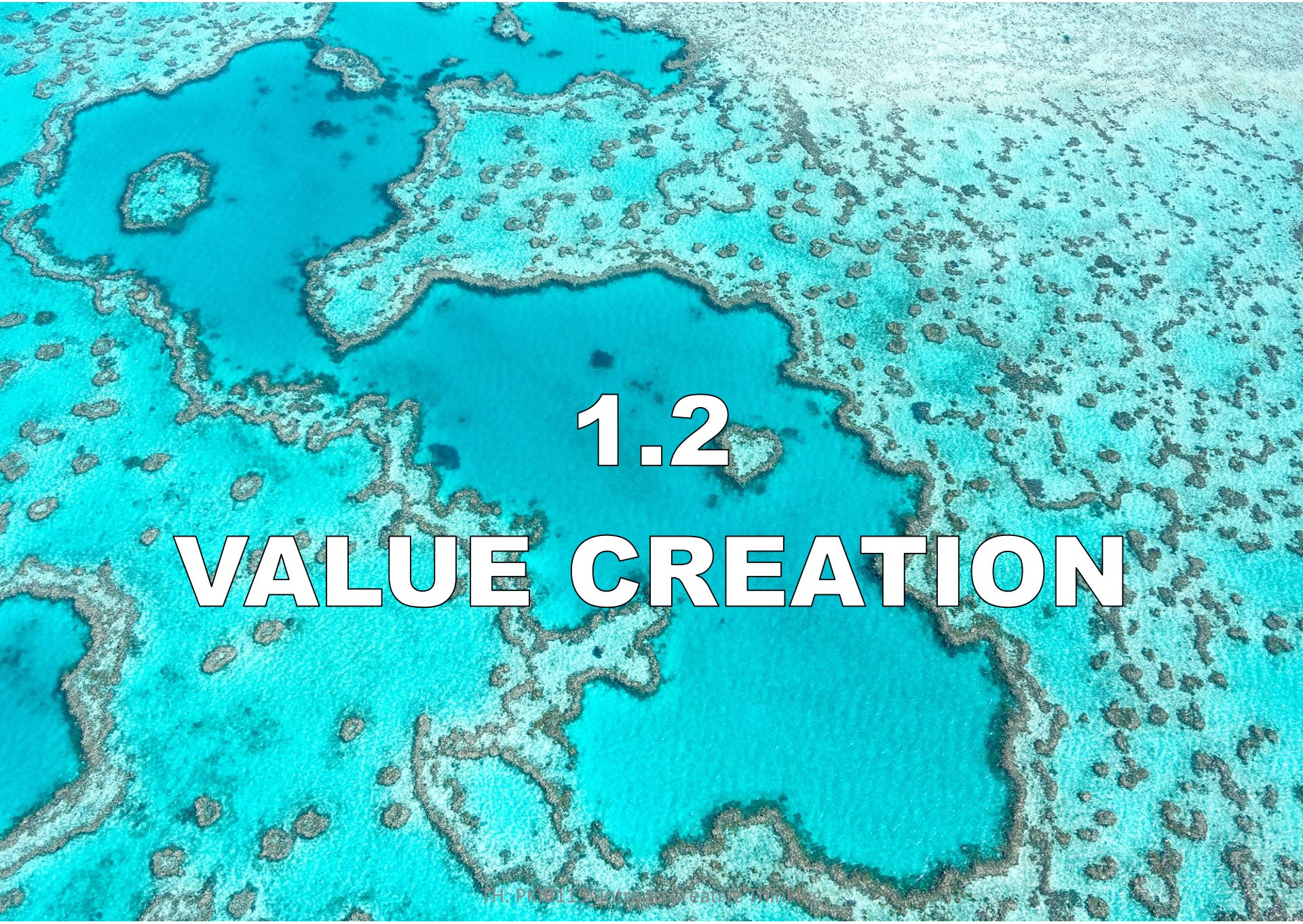
4 MODELS TO MEASURE CREATIVITY

- According researchers, measuring creativity largely depends on one's **purpose** of looking at it.
- There are 4 common models to measure creativity:

1. The Csikszentmihalyi Systems Model – *social acceptance*
2. The Taxonomy of Creative Designs – *level of creativity*
3. The Guilford Measures – *divergent thinking*
4. The Requirements Model – *specific field of study*

Source: Nilsson (2011); Stana (2017)

ALL THE ABOVE LEAD TO VALUE CREATION


1.2
VALUE CREATION

1. THE CSIKSZENTMIHALYI SYSTEMS MODEL

- introduced by Mihalyi Csikszentmihalyi
- to measure how creative a **piece of work** is within its **social setting**:
- to do with appreciation of work between:
  - ✓ the **artist**
  - ✓ the **domain** (of expertise)
  - ✓ the **field** (of community)



Source: Nilsson, Peter (2011)

VALUE CREATION IN ART



"The Embrace" by Sergey Gorshkov, winner of Wildlife Photo of the Year 2021 – techradar.com



Water Lillies by Claude Monet – dw.com
- Sold for $47.3 million



"Circular Octopus" by Alessandro Grasso



Les Voyageurs by *Bruno Catalano* – interestingengineering.com

VALUE CREATION IN DESIGN





Habitat 67, Montreal Canada by Moshe Safdie





The Bubble House, by Antti Lovag - twistedsifter.com

AESTHETIC SENSIBILITY

Edward de Bono, professor in lateral thinking stated,

*"In the art world, creativity involves **aesthetic** sensibility, **emotional** resonance and **a gift for expression**"*

De Bono, E. (1970). Lateral Thinking: Creativity Step-by-Step. p.11

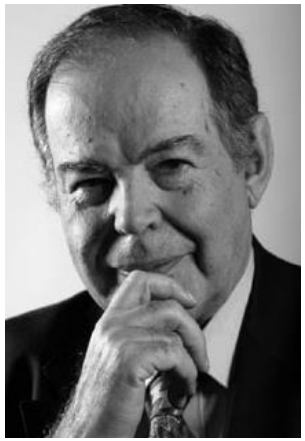


Image: edwarddebonofoundation

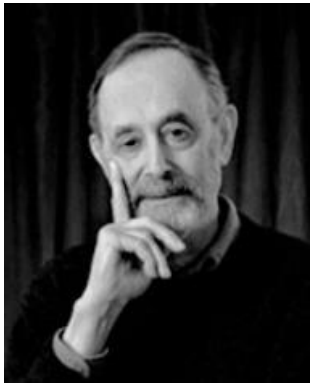


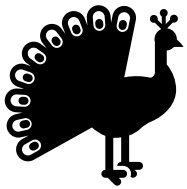
Image: arnoldberleant.com

Arnold Berleant, professor of Philosophy and Music said,

*"Aesthetics is a theory of **emotions** defined by … **a cultivated sensibility**…"*

*Berleant, A. (2015). "*Aesthetic Sensibility"*. Ambiences. DOI:10.4000/ambiances.526*

ELEMENT OF VALUE CREATION #1



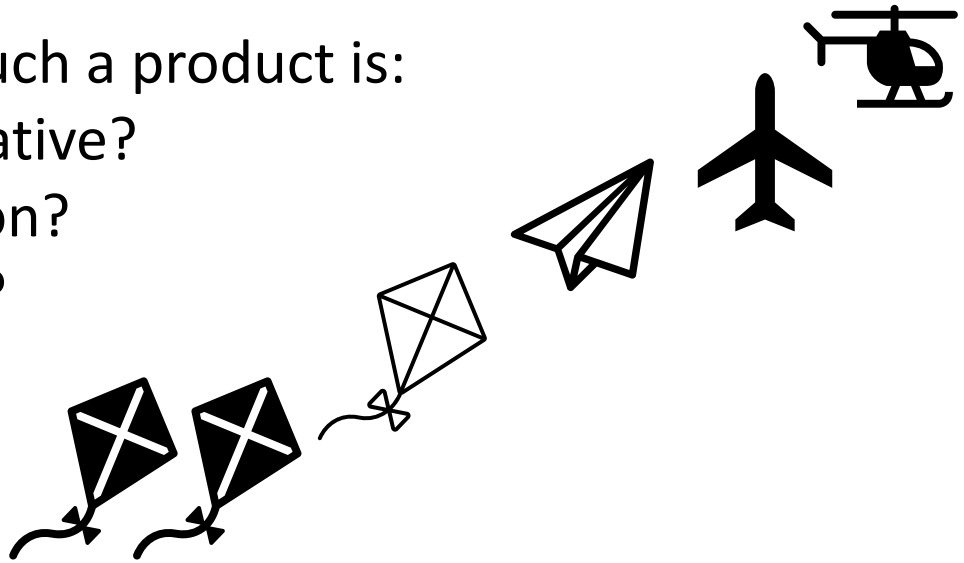
Attractive

- What is beautiful?
- Pleasing to the eye
- How does it make you feel?
- **Emotions** that it evokes
- **Subjective** to one's perspective
- Relates to **gifted** talent

2. THE TAXONOMY OF CREATIVE DESIGNS

- to measure how creative **a piece of work** is by **degree of originality**

- It is to see how much a product is:
  - ✓ original/ innovative?
  - ✓ a transformation?
  - ✓ a combination?
  - ✓ a variation?
  - ✓ an imitation?



  - The degree of originality ranges from reproduction to innovation

Source: Nilsson (2011); Stana (2017)

ELEMENT OF VALUE CREATION #2



Novel:

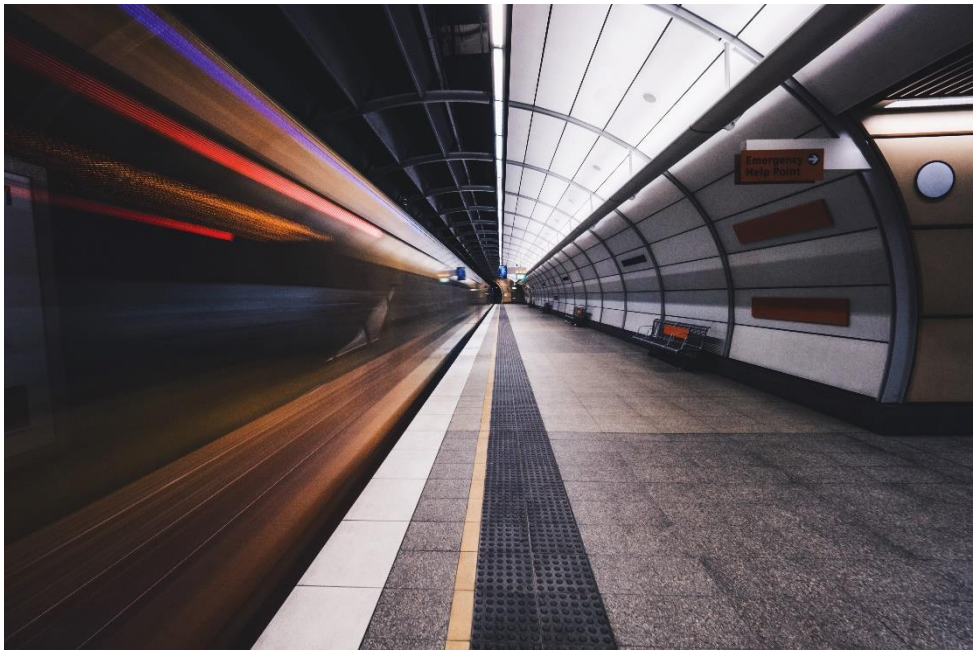
original, new

→ unexpected

Source: Sternberg & Lubart (1999) in Pfiffer, D2.(2012)

3. THE GUILFORD MEASURES

- Introduced by Joy Paul Guilford and Ellis Paul Torrance
- to measure how creative **a person** by looking at the person's ability to do **divergent thinking** and to come up with **ideas that go in many different directions**
- How far can you move away from the mainstream idea?
- Divergent thinking can be trained



Source: Nilsson, Peter (2011)

SCHOLARS' DEFINITION

Robert J. Sternberg, Professor of Human Development at Cornell University said,

*" (Creativity is)… the creation of **new** and **useful** products including ideas as well as concrete objects"*

(Meyers in Sternberg, R. 1999. *Handbook of Creativity*)



Image: famouspsychologists.org



Image: issci.online

Todd I. Lubart, Professor of Psychology at University of Paris Descartes stated that,

*"Creativity from the Western perspective can be defined as the ability to produce work that is **novel** and **appropriate**"*

(Lubart, T.I. 1999: 339 in Sternberg. *Handbook of Creativity* at https://creativity.netslova.ru)

SCHOLARS' DEFINITION

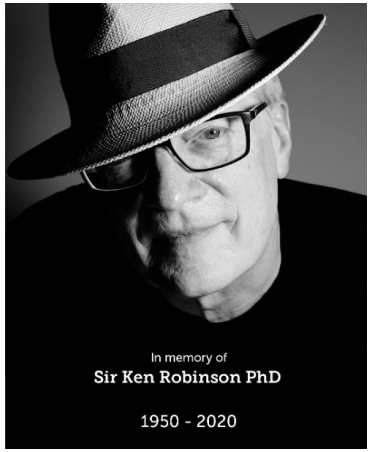


Image: sirkenrobinson.com

Ken Robinson, Professor of Arts Education at University of Warwick, England defines it as,

*"… the process of having **original ideas** that **have value**"*

(Robinson, K. 2006, *Ted Talk*)

Edward de Bono, Da Vinci professor of Thinking Chair at the University of Advancing Technology, Arizona highlights,

*"… one of the main purposes for the use of creative thinking is **to find better ways of doing things**"*

(De Bono, E. 1993, *Serious Creativity*)

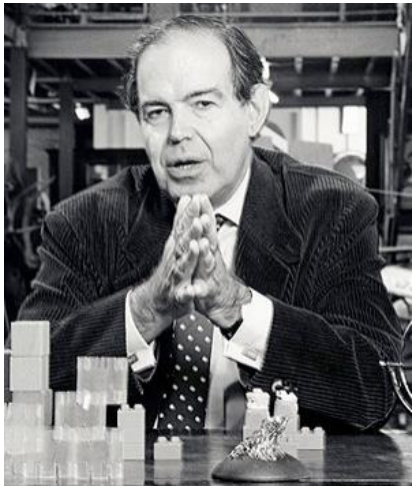
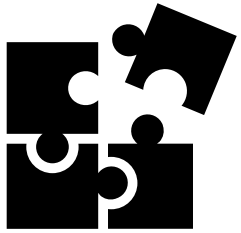


Image: en.wikipedia.org

ELEMENT OF VALUE CREATION #3

Appropriate: 

- to identify the most suitable solution among many

- Most useful, has value, offers better ways of doing things

→ fitting, relevant

(Runco and Albert, 1990; Runco and Pritzke, 1999; Sternberg 1999 in creativity.netslova.co.uk)

4. THE REQUIREMENTS MODEL

- to measure how creative a **piece of work** is against a programme or a field:

- e.g. painting
- e.g. photography
- e.g. architecture
- e.g. engineering
- e.g. interior design

- uses **strict criteria** within the programme
- criteria is defined by the **community of experts** in the field

Museum of the Future, Dubai





- Opened in 2022

Intro to museum

- https://www.youtube.com/watch?v=eR8K-lCYVNk&ab_channel=NasDaily

ELEMENT OF VALUE CREATION #4



Social impact

- accepted by the **community**
- **solves society's problems**
- improves economy
- makes life richer, more interesting

4 MODELS TO MEASURE CREATIVITY

Measuring creativity depends on one's **purpose**:

1. *The Csikszentmihalyi Systems Model* (Mihalyi Czikszentmihalyi)
   - to measure how creative a piece of work is within its social setting
   →**ATTRACTIVE**

2. *The Taxonomy of Creative Designs*
   - to measure how creative a piece of work is
   → **NOVEL**

3. *The Guilford Measures (Joy Paul Guilford, Ellis Paul Torrance)*
   - to measure how creative a person is (divergent ideas) → **APPROPRIATE**

4. *The Requirements Model*
   - to measure how creative a piece of work is against a programme (using strict criteria) → **SOCIAL IMPACT**

Source: Nilsson, Peter (2011)

CREATIVITY AND VALUE CREATION = N.A.S.A



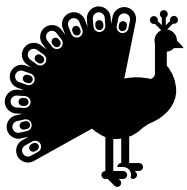
Novel



Social impact



Appropriate



Attractive

VALUE CREATION is to do with

INTERACTION BETWEEN HUMAN AND THE ENVIRONMENT

VALUE CREATION IS A DYNAMIC CONCEPT

- Value creation in the **past** is related to **economies of industrial scale**:

  mass production, efficient repeatable tasks

- Value creation **today** is based more and **more on creativity**:

  innovative ways to utilise new materials, technologies, and processes

- Value creation in the **future** will be based on **economies of creativity**:

  mass customization for service improvement to market, finding solutions, new ways to sell and **deliver to customers**



Source:
Hughes, J (2013) "What Value Creation Look Like in the Future". Harvard Business Review. https://www.hbr.org



Image: Gareth Cattermole/ Getty in businessinsider.com

AIM OF THE COURSE

- To develop you as a creative person
- so that you can develop creative ideas, concepts, products, systems, solutions, etc.
- to cope with the changing world

CREATIVE VS CRAZY

"… every valuable creative idea must always be **logical** in hindsight. If an idea were not logical in hindsight then we would have no way of seeing the value of that idea and it would simply be a "crazy" idea."

(De Bono, 1993; 2015. *Serious Creativity*. Vermillion, London)



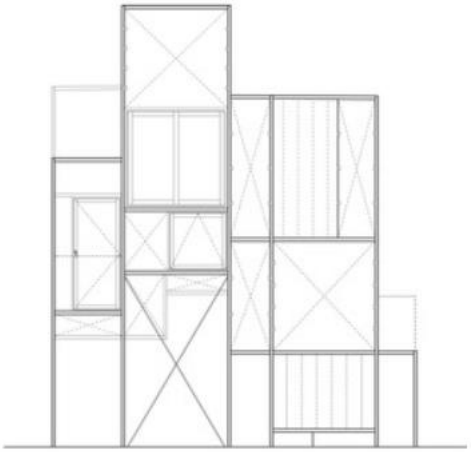
Image: pulptastic.com

RATE THIS INNOVATION BASED ON N.A.S.A.

House NA, Japan



archdaily.com



What if I tell you that it embodies the concept of "unity in separation"



RATE THESE INNOVATIONS BASED ON N.A.S.A

Head-mounted tissue paper



Image: wonderslist.com

Multi-pour kettle





Remote control holder

Splash control



Image: wonderslist.com

Imagefacebook.com//MyPublicitInc

RATE THESE INNOVATIONS BASED ON N.A.S.A.

Bald-head crash helmet





Image: express.co.uk

The Hoverella



Image: pocket-lint.com

Noodle-cooling fan chopstick



Imagefacebook.com//MyPublicitInc

PADLET WORK #2

Choose ONE innovation; improve its design

Upload your improved products on PADLET

1.3

SIMPLE AND COMPLEX DEFINITION

This is a typescript of an early version of Cropley, A. J. (2011). Definitions of creativity. In M. A. Runco & S. R. Pritzker (Eds.), *Encyclopedia of creativity* (pp. 511-524). San Diego, CA: Academic Press.

DEFINITION

Arthur J. Cropley
University of Hamburg

Novel

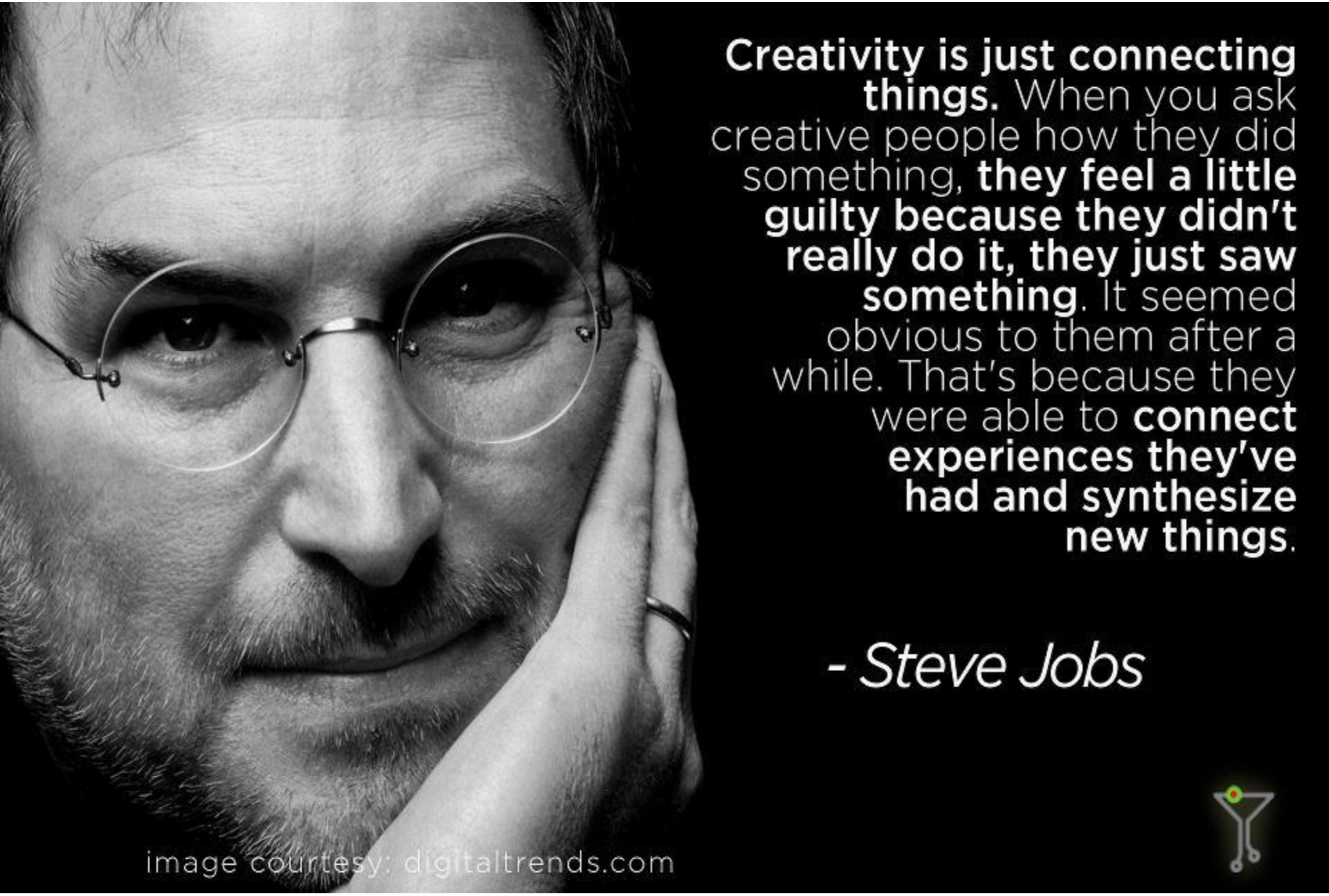
Appropriate

Social impact

Attractive

Summary

Although the modern definition of creativity has moved away from esthetics and discovery towards emphasis on meeting competition, the idea of novelty is central (although not necessarily sufficient). Also necessary are relevance and effectiveness, as well as ethicality. Novelty is understood in different ways, and this leads to a distinction between creativity in the sublime and in the everyday sense. Although both creativity and intelligence require knowledge and effort, they can be distinguished from each other, and much the same can be said about creativity and problem solving. Creativity can also be defined as a social phenomenon that is facilitated by some social factors, inhibited by others. One important social setting is the place of work, where an interaction between the person and the environment affects the process of innovation. Focusing on the individual person, creativity is defined as an aspect of thinking, as a personality constellation, and as an interaction between thinking, personal properties and motivation. This interaction involves a number of paradoxes, in that apparently contradictory elements have to coexist for creativity to emerge.


Creativity is just connecting things. When you ask creative people how they did something, **they feel a little guilty because they didn't really do it, they just saw something**. It seemed obvious to them after a while. That's because they were able to **connect experiences they've had and synthesize new things**.
- *Steve Jobs*
image courtesy: digitaltrends.com

DO YOU AGREE WITH STEVE JOBS THAT IT IS JUST ABOUT MAKING CONNECTIONS?

“They didn’t really do it”

“They just saw something”

SIMPLE DEFINITION



"Creativity is about *MAKING CONNECTIONS* where none existed before."

- Yew Kam Heong, 2013

SIMPLE DEFINITION

It begins from a simple concept of

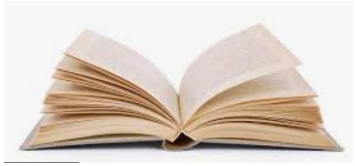
A + B = C

MAKING CONNECTIONS



Image: johnlewis.com

+



=

MAKING CONNECTIONS



Image:lacrema.com

+



Image:istock.com

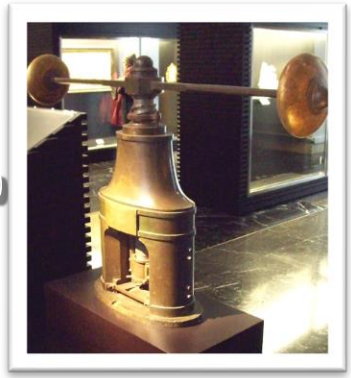
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JOHANNES GUTENBERG



Wine press

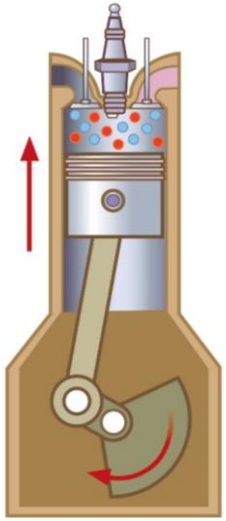
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Coin punch

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OLE EVINRUDE



Petrol engine

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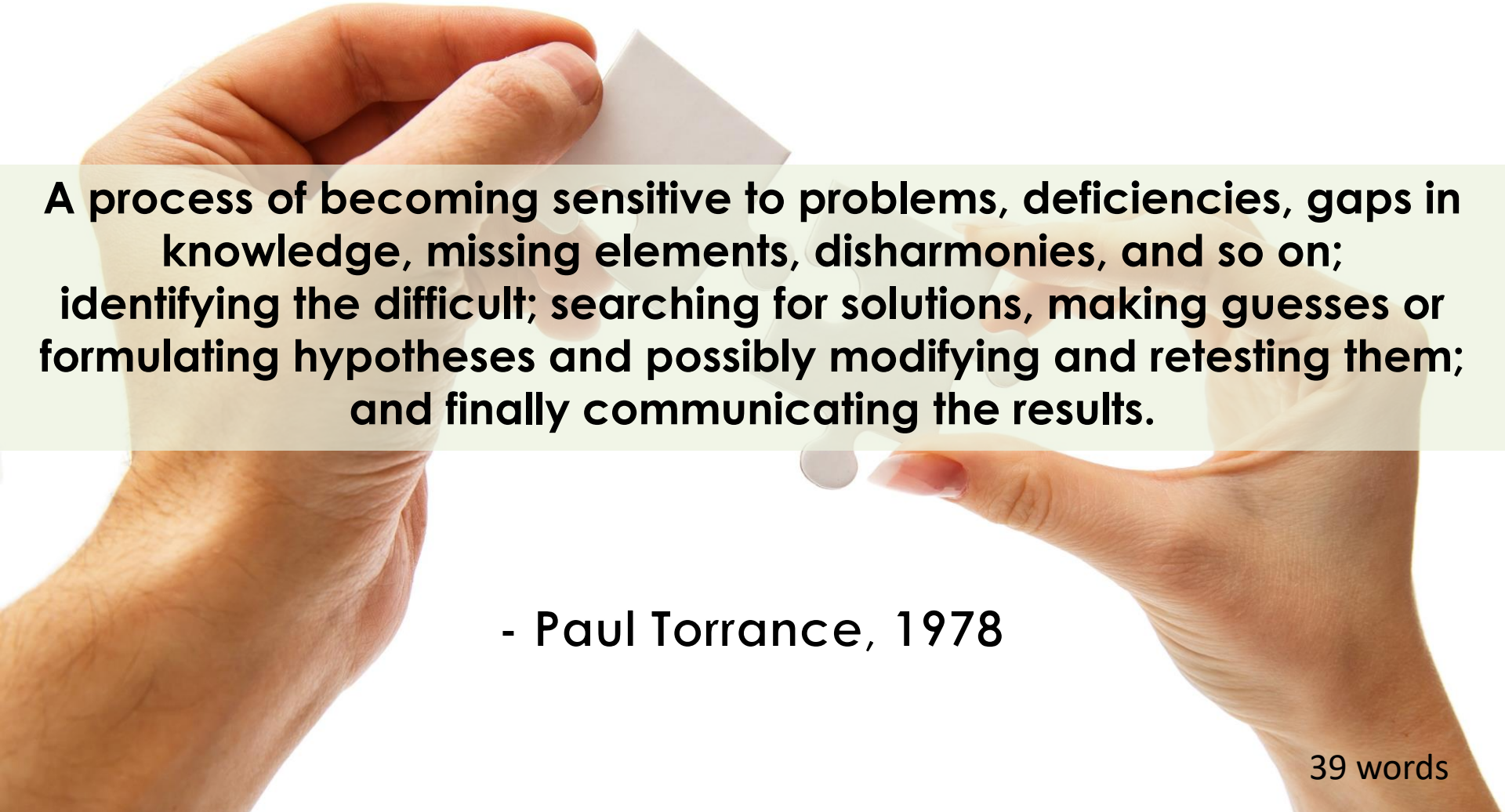


Paddling boat

=

WHAT OTHER CREATIVITY DO YOU SEE AROUND YOU THAT IS A FUNCTION OF MAKING CONNECTIONS?

COMPLEX DEFINITION



A process of becoming sensitive to problems, deficiencies, gaps in knowledge, missing elements, disharmonies, and so on; identifying the difficult; searching for solutions, making guesses or formulating hypotheses and possibly modifying and retesting them; and finally communicating the results.

- Paul Torrance, 1978

39 words

MAKING CONNECTIONS

A process of becoming

…sensitive to problems, deficiencies, gaps in knowledge, missing elements, disharmonies, and so on;
identifying the difficult;

…searching for solutions, making guesses or formulating hypotheses and possibly modifying and retesting them;

…finally communicating the results.

~ Torrance, 1974

① **Creativity involves a process:**

② **Identify the problem or the need to improve**

③ **Solve the problems**

④ **Share the new idea, commercialise the solution**

MAKING EXTENDED CONNECTIONS

The connection can be expanded from

A + B + C + D ... = Z

BICYCLE

1817+ hobby-horse, bone-shaker

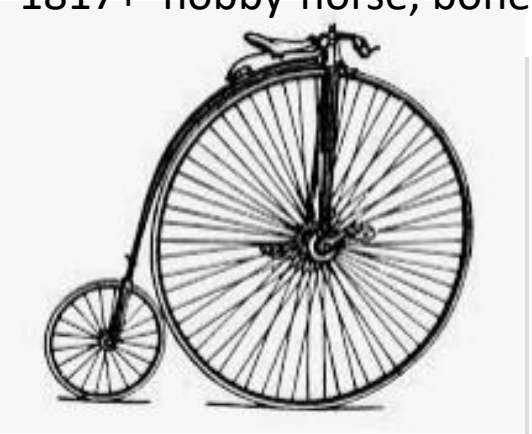


Image: antiques.lovetoknow.com





Image: precisiontandems.com on pinterest.com



Image: nymag.com



Image: bicycling.com



Image: bikeradar.com

MAKING EXTENDED CONNECTIONS

Products/ services
Touch + go
Drive thru' food
Grab

Discuss.
Share your findings




1.4
BIG C
AND SMALL c

THE FOUR-C MODEL OF CREATIVITY

The 4 C's of Creativity:

- The big "C" (genius C)
- The pro-"C" (expert C)
- The little "c" (personal c)
- The mini "c" (everyday c)

(Kauffman & Beghetto, 2009)



Image: medium.com

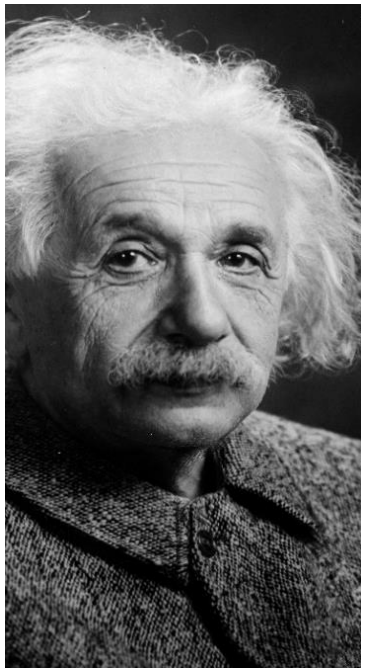


Image: theguardian.com



Image: facebook.com/stephenhawking

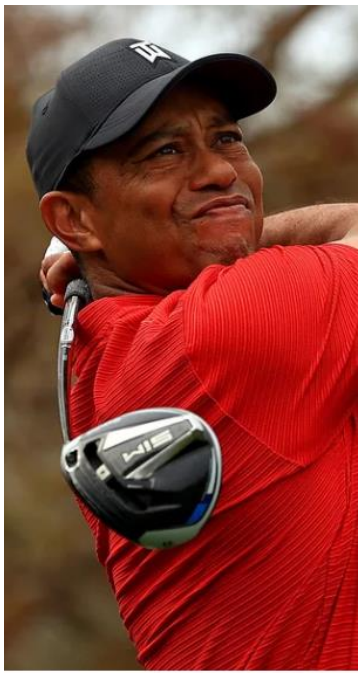


Image: npr.org



Image: facebook.com/realstanlee

THE BIG C & THE PRO-C

THE LITTLE-C AND THE MINI-C

Creative expressions



Image: Anna Abramskaya on pinterest.com



Image: pexels.com

Everyday problems



Image: pexels.com

You need to make payment for fuel but your phone is dead. You have not got your power bank or your purse with you for other means of payment.



Image: flightcalculator.com

You are late by 15 minutes at the airport check-in counter. There are no airline staff available. Customer service can't help you since you have not checked-in online and you don't have the boarding pass. You're about to miss your flight

Everyone has **creative potentials**. You need to bring it out. It can be learned and trained.

CHAPTER SUMMARY

- Creativity as a limitless concept
- Value Creation
- Simple and complex definition
- Big C and small c

PADLET WORK #1

- Introduce yourself on PADLET



SEE YOU IN THE NEXT CLASS

